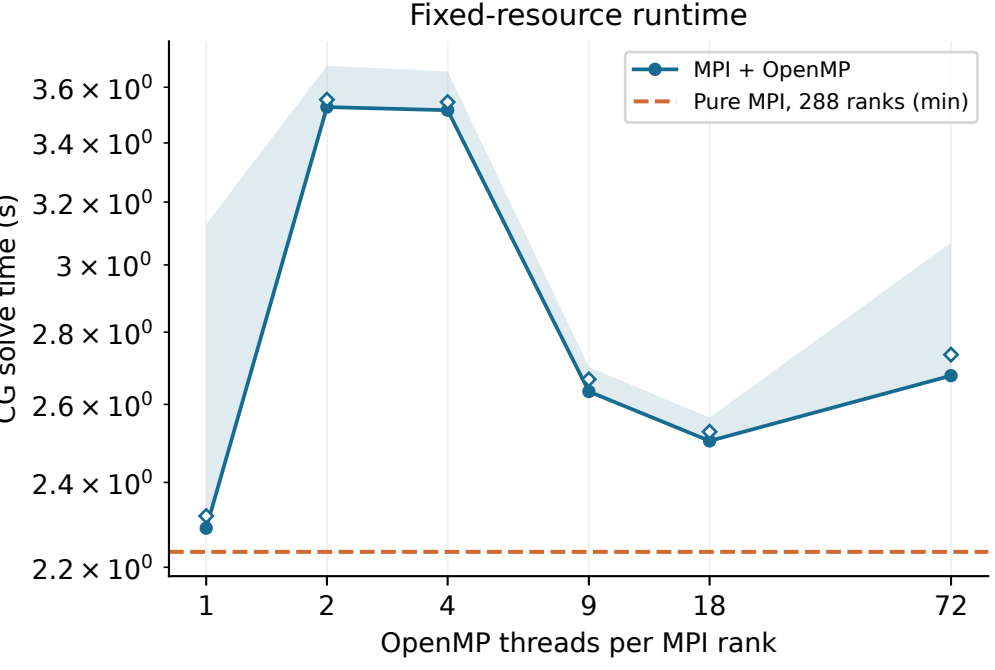
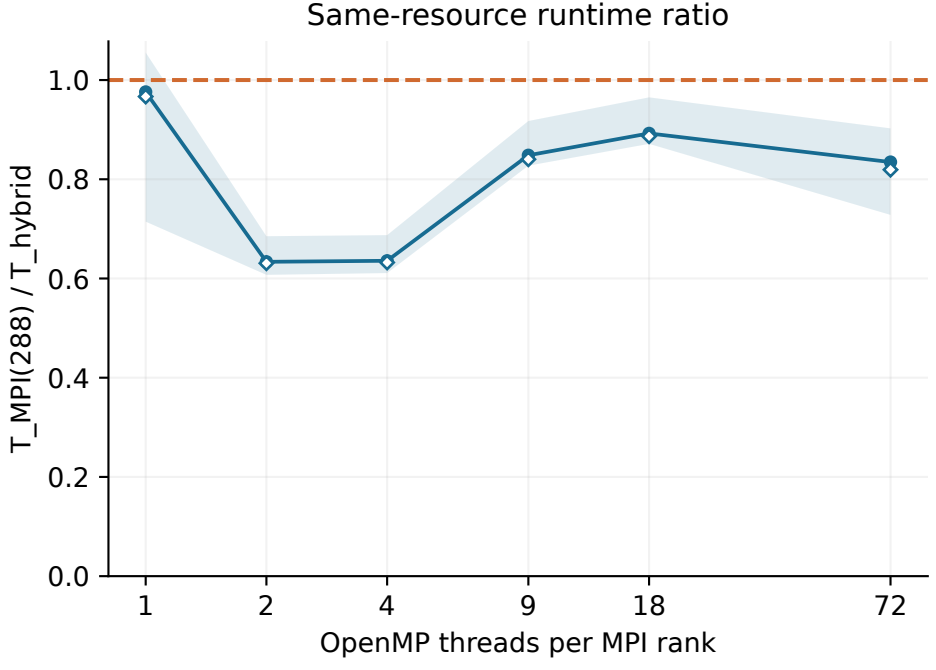
2D MPI + OpenMP | n=2048 | fixed 288 physical cores

Vanda | audited CPU measurements | 3 independent runs per configuration





Line/filled marker: minimum; hollow diamond: median; shading: observed min-max (3 runs, not a CI).
MPI ranks x threads: 288x1, 144x2, 72x4, 32x9, 16x18, 4x72. Ratio is not a strong-scaling efficiency.